Weekly Progress Report

Project Title: Quantum Secure Email Client Application (QuMail)
Team ID: G14
Date: Week 1

Executive Summary

Current Sprint: Sprint 1 of 6

Sprint Goal:
The main goal of the first sprint was to initiate the project by understanding and finalizing the problem statement, identifying the major requirements of the Quantum Secure Email Client Application, and setting up the basic project management and development environment.

Status: On Track

During this week, the project problem statement and its scope were discussed and finalized. The main objective is to develop an email client application, named **QuMail**, which can integrate Quantum Key Distribution (QKD) with existing email communication. The application is intended to provide different levels of security while maintaining compatibility with commonly used email services and protocols.

The initial project planning was also started. The basic Jira structure and product backlog were prepared to organize the upcoming development activities. A Git repository was also initialized so that the project source code and future changes can be maintained properly.

The major features, security levels, system components, and expected users of the application were identified during this week. Based on this understanding, the initial requirements and development tasks were added to the backlog.

UML & OOAD Progress

Diagrams Completed this week:
No major UML diagrams were completed during Week 1. The week was mainly focused on project initiation, requirement understanding, and defining the overall scope of the system.

Diagrams Revised:
No diagrams were revised during this week.

OO Concept Applied:
The basic modular structure of the application was identified. The system will be divided into separate modules such as the User Interface, Email Service, Key Manager interface,

Encryption/Security module, and Email Processing module. This will make the application easier to maintain and allow individual modules to be upgraded in the future.

The project will follow an object-oriented approach so that different responsibilities of the system can be handled through separate classes and components.

Technical Execution (DevOps & Git)

Jira Tasks Completed:

- Project board was created for managing the development activities.
- Initial product backlog was prepared.
- Major project requirements were converted into initial development tasks.
- Initial sprint activities were planned.
- Tasks related to email integration, QKD/Key Manager integration, encryption levels, GUI, and testing were identified for future sprints.

Git Activity:

- Git repository was initialized for the project.
- Initial project structure was planned for maintaining the source code.
- The repository will be used for version control and collaborative development during the upcoming sprints.

CI/CD Status:
CI/CD pipeline has not been implemented in Week 1. It will be considered during the development phase after the initial application structure and codebase are created.

Testing Summary

Total Unit Tests: 0

Pass Rate: N/A

Key Bug Identified:
No implementation-related bugs were identified during Week 1 because development and unit testing have not started yet.

The main challenge during this week was understanding how QKD-based key management can be incorporated into a conventional email client without changing the basic email communication workflow. The initial approach is to introduce the encryption mechanism at the application layer so that the email can continue to use existing email services while the security layer handles the encryption and decryption process.

Goals for Next Week

- Finalize the detailed functional and non-functional requirements of the QuMail application.
- Prepare the initial UML diagrams such as Use Case Diagram and Class Diagram.
- Define the overall system architecture and identify the interaction between the User, QuMail, Key Manager, and Email Server.
- Finalize the initial database/data requirements, if required for the application.
- Start designing the GUI structure of the email client.
- Study the integration requirements of the Key Manager API and existing email protocols.
- Break down the major features into smaller Jira tasks for the next sprint.
- Continue maintaining the project source code and documentation through Git.

Project Guide Signature **Project In-charge Signature**